Quran-Tajzee: Balanced Quran Divisions that End Where a Thought Ends

Tarek Eldeeb
Giza, Egypt
tarekeldeeb@gmail.com

***Abstract*—The Madina Mushaf divides the Quran into 30 juz, 60 hizb and 240 quarters (rub^{c}), taken from al-Ṣafāqusī's *Ghayth al-Naf^{c}*. His stated rule is that each quarter ends at a complete stop, at a position that is agreed or well known; the quarters are not computed. We measure the transmitted divisions on the Tanzil text with two length models, Hafs recitation in harakat with madd lengths and DigitalKhatt print width, which correlate at $r = 0.990$ per aya. The juz lie within 2.4% of equal-letter positions, but the quarters lie 10× further from equal positions than aya ends force and vary 2.2× in length; and 90 of 239 boundaries fall inside a subject phrase. Keeping the traditional juz, we re-cut hizb and quarters by an exact dynamic program restricted to the starts of 1,112 reviewed subject phrases of 1–10 ayat. The worst hizb drops from ±13.9% to ±3.6% of half a juz (CV 6.9% → 1.4%), and no new boundary falls inside a phrase. Quarter balance is limited by phrase granularity; we show that about a hundred targeted, text-checked phrase splits bring the quarter CV from 11.7% to near the aya-level floor of 2.0%.**

***Index Terms*—Quran, text segmentation, partitioning, dynamic programming, typesetting, recitation, tajweed**

I. Introduction

Readers, teachers and printers organise the Quran by its divisions: a juz a day completes it in a month, a hizb or a quarter is a daily portion or a unit of revision, and printed Mushafs mark every quarter. Those divisions are of scholarly judgement (*ijtihādī*), not revelation, which is why mushafs and regions differ on them [3]. The Madina Mushaf takes its 30 juz, 60 hizb and their quarters from *Ghayth al-Naf^{c}* by al-Ṣafāqusī (d. 1118 AH) [1], [2].

Two properties are wanted of a division: units of similar length, and units that are complete in meaning. The juz satisfy the first; the quarters satisfy neither fully (Section II). This paper (i) measures the traditional divisions against their own stated method, (ii) defines a length model for recitation and print, (iii) solves the hizb and quarters inside the traditional juz exactly, with every boundary at the start of a subject phrase, and (iv) shows how targeted phrase splits, each checked against the text, reach the balance that aya granularity allows. Data, code and every number in this paper are in the open repository [13].

II. The traditional method and its outcome

In the terminology section of his introduction al-Ṣafāqusī states his method [1]: he does not compute boundaries but follows only the agreed or well-known ones, "ولا أمشي إلا على المتفق عليه أو المشهور"; he treats each quarter as a unit, "وأذكر حكم كل ربع بانفراده"; and he closes each on a proper stop, naming its last word, the kind of stop it allows and whether it is an aya end. At 2:25 he writes "خالدون تام في أعلى درجاته وفاصلة ومنتهى الربع بإجماع": a complete stop of the highest degree, an aya end, the end of the quarter by consensus. Where traditions differ he lists the positions of the Maghāriba and the Mashāriqa and states a preference; the two cases we checked both match the Madina Mushaf (45:12, 82:1).

Drawn at stops, not by count. The early divisions were made by counting letters [4], [5]. Counting the 325,386 written letters of the text (basmala excluded), the traditional juz start on average 2.4% of a juz from equal-letter positions. Inside each juz, however, the quarters start 15% of a quarter from equal positions, while the nearest aya end is only 1.5% away: 10× further than aya granularity forces. Quarter lengths vary 2.2× in letters. The quarters were placed at the ends of thoughts, as the method says, not by count.

But not always at the ends of thoughts. We segmented the text into subject phrases (Section III). Against that reading, 90 of the 239 traditional boundaries, 79 of the 210 hizb and quarter boundaries inside the juz, fall inside a phrase, at a median 50% of the way through it. Some open mid-dialogue or mid-story: juz 16 at "قال ألم أقل لك" (18:75), juz 27 at "قال فما خطبكم أيها المرسلون" (51:31), a hizb at "فنبذناه بالعراء" (37:145). The quarters thus pay the price of uneven length without fully keeping the rule they were drawn by. This is the problem we address.

III. Data and length model

Text. Tanzil Uthmani 1.1, 6,236 ayat in the Kufan count [8].

Subject phrases. The text was divided into runs of 1–10 consecutive ayat on one subject, drafted by reading the text and reviewed by hand: 1,094 phrases as a reading of the text, 1,112 after the review round of Section IV. A phrase start is our proxy for a complete stop: the next aya does not continue the previous subject.

Recitation size. Per aya, the phonemes of the Hafs recitation from *tajweed-embeddings* [10]: a madd letter counts its length in harakat (2, 4 or 6, with *tawassuṭ* for the optional lengths), a silent letter 0, every other letter 1. Each aya is recited continuously and stopped at its end. A surah opening adds the basmala (19 harakat), except At-Tawba. The whole text is 398,670 harakat.

Print size. Per aya, the width of the aya set as a single line in the DigitalKhatt Madina font [9], shaped with HarfBuzz by Quran-Nazm [11], with its aya marker and one space.

The two sizes correlate at $r = 0.990$ per aya and $r = 0.994$ per phrase. We mix them 50/50 as shares of their totals, $c_a = \frac{1}{2} r_a/R + \frac{1}{2} p_a/P$.

IV. Method

Keep the juz. An exact solver over the phrases (penalising mid-surah starts) can make the 30 juz more even, but only from ±6.2% to ±4.9%. The traditional juz are kept; the imbalance is in the finer divisions.

Nested exact cuts. Each juz is cut into two hizb, then each hizb into four quarters, the way the divisions nest. A container G of ayat cut into k parts has target $T = |G|/k$ and the admissible cuts are phrase starts $\mathcal{S}$. Lexicographically we

$$\min_{\text{cuts}\subset\mathcal{S}} \max_q \frac{|S_q - T|}{T}, \quad \text{then} \quad \min \sum_q (S_q - T)^2, \qquad (1)$$

both by the dynamic program $F_k(i) = \min_{j<i} [F_{k-1}(j) \oplus g(C_i - C_j)]$ over prefix sums C, with $\oplus = \max$ or $+$ [12]. Contiguous partitioning has optimal substructure, so the solution is the global optimum; tests check it against exhaustive search. A *penalized* variant also admits any aya end at a cost m^2 per mid-phrase start; we use it only to *propose* phrase splits.

Review loop. Balance is limited by how fine the phrases are. For each container, the aya end that best balances it is computed; if it lies inside a phrase, the text at it and at the other candidates closer to balance is read, and the phrase is split only at a complete stop opening a new ruling, scene or subject; otherwise the current start is kept. The split becomes a phrase boundary, so the rule "every boundary opens a phrase" is never relaxed.

V. Results

Hizb. One review round checked all 30 second-hizb starts: 10 were already at the best point, 7 phrases were split at complete stops (e.g. 9:30 "وقالت اليهود", 10:28 "ويوم نحشرهم"), and 13 were kept because every closer point lies inside a story or dialogue (e.g. juz 6, +3.6%, inside the dialogue of Mūsā and his people). The 11 phrases cut by traditional juz starts were split too. Table I gives the balance.

Quarters. With phrase starts only, a hizb holds about 18 phrases, 4.6 per quarter, and one long phrase can be a fifth of a quarter; the worst quarter improves only to ±23.8%. Fig. 1 shows what targeted splits buy: the penalized solver proposes the splits, the phrase-only solver is re-run with them accepted. About a hundred splits bring the quarter CV close to the floor of 2.0% reached when every aya end is allowed; the remaining worst case, ±7.5%, is set by single long ayat (2:282 alone is 40% of a quarter), and going lower would require cutting inside an aya, which no traditional division does. Uniform re-segmentation is far less efficient: splitting every phrase to ≤3 ayat adds about 1,350 boundaries for a CV of 4.5%.

Table I: Traditional and new divisions (50/50 size mix). Hizb against $\frac{1}{2}$ of their juz, quarters against $\frac{1}{4}$ of their hizb; ratios over the whole Quran.

	Traditional	Quran-Tajzee
Worst hizb	±13.9%	±3.6%
Hizb CV	6.9%	1.4%
Longest/shortest hizb	1.36×	1.17×
Worst quarter	±38.3%	±23.8%
Quarter CV	11.7%	9.2%
Longest/shortest quarter	2.19×	1.65×
Starts inside a phrase (of 210)	79	0

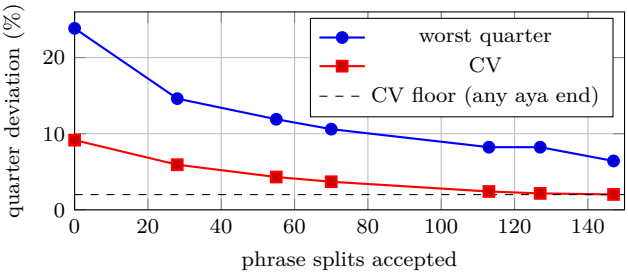


Figure 1: Quarter balance against the number of targeted phrase splits.

VI. Discussion and limitations

The phrase test is ours. That no new boundary falls inside a phrase holds by construction; the phrases are one reading of the text, under review. A wording check independent of it, a first word that leans on the previous aya (a reply such as "قال" or a linking "ف", "ثم"), flags 21 traditional starts and a similar number of new ones, most of them a resumptive "ف" at a new subject ("فاصدع بما تؤمر"), but a few real: the 10-aya phrase limit splits long stories at turns of dialogue. An independent judge is the pause grades of the classical *waqf* works, al-Dānī's *al-Muktafā* and al-Ushmūnī's *Manār al-Hudā* [6], [7], at every candidate aya end.

Counting schools. Boundaries are Kufan aya ends; al-Ṣafāqusī also requires an aya end agreed by all counts (فاصلة بلا خلاف), not yet checked. **Weights.** Recitation and print agree so closely that the 50/50 mix is not sensitive; header lines of surahs, if counted in print, lengthen juz 30.

VII. Conclusion

Al-Ṣafāqusī's rule, that a quarter ends where a thought ends, is sound, but the transmitted quarters keep it only partly and vary more than twofold in length. Applying the rule to every boundary and choosing among the admissible stops by measured length gives hizb within ±3.6% of half their juz, none inside a phrase. For the quarters, the next review round will split about a hundred phrases at checked stops, which brings their spread close to what aya granularity allows.

References

[1] ᶜA. al-Nūrī al-Ṣafāqusī, *Ghayth al-Nafᶜ fī al-Qirāʾāt al-Sabᶜ*, introduction and passim.

[2] King Fahd Complex for the Printing of the Holy Quran, *Muṣḥaf al-Madīna* (Ḥafṣ), closing note on its sources.

[3] Gh. Q. al-Ḥamad, *Taḥzīb al-Qurʾān fī al-Maṣādir wa-l-Maṣāḥif*.

[4] Ibn Abī Dāwūd, *Kitāb al-Maṣāḥif*.

[5] Abū ᶜAmr al-Dānī, *al-Bayān fī ᶜAdd Āy al-Qurʾān*.

[6] Abū ᶜAmr al-Dānī, *al-Muktafā fī al-Waqf wa-l-Ibtidāʾ*.

[7] A. al-Ushmūnī, *Manār al-Hudā fī Bayān al-Waqf wa-l-Ibtidāʾ*.

[8] Tanzil Project, "Quran text, Uthmani, version 1.1," https://tanzil.net.

[9] A. Anane, "DigitalKhatt," https://digitalkhatt.org.

[10] T. Eldeeb, "tajweed-embeddings," https://github.com/tarekeldeeb/tajweed-embeddings.

[11] T. Eldeeb, "Quran-Nazm," https://github.com/tarekeldeeb/quran-nazm.

[12] R. Bellman, "On the approximation of curves by line segments using dynamic programming," *Commun. ACM*, vol. 4, no. 6, p. 284, 1961.

[13] T. Eldeeb, "Quran-Tajzee," https://github.com/tarekeldeeb/quran-tajzee.